

10.3 Matrix Representations of Graphs

10.3.1 Example 10.3.19

$$\text{Let } A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 6 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 5 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 6 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 5 \end{bmatrix} = \begin{bmatrix} 15 & 9 & 15 \\ 9 & 5 & 8 \\ 15 & 8 & 8 \end{bmatrix}$$

Assume A is an adjacency matrix. How many walks of length 2 from v_1 to v_3 ? 3.

Assume A is an adjacency matrix. How many walks of length 3 from v_1 to v_3 ? 15.

Examine the calculations you performed to find five walks of length 2 from v_3 to v_3 . Then draw G and find the walks by visual inspection.

$$2 \cdot 2 + 1 \cdot 1 + 0 \cdot 0 \quad (1)$$

Left edges from v_3 to v_1 , right is edges from v_1 to v_3 . $2 \times 2 =$ walk of length 2 from v_3 to v_3 via v_1 .

10.2 Matrix Representations of Graphs

Graph in book.

- How many from a to c ? **4**.
- How many trails from a to c ? **4 + 3 + 2 + 1**
- How many walks from a to c ? ∞

10.2.1 Example 10.3.6b

An edge whose removal disconnects graph it is part of is called an bridge. Find the bridge in:

The graph is in the book

- $\langle v_7, v_8 \rangle$
- $\langle v_3, v_4 \rangle$
- $\langle v_1, v_2 \rangle$

10.2.2 Example 10.2.8d

Find the number of connected components in: *Image in book.* **2.**

10.2.3 Euler Circuit

Let G be a graph. An Euler circuit for G is a circuit that contains every vertex and every every edge of G . That is, an Euler Circuit for G is a sequence of adjacent vertices and edges in G that has at least one edge, starts and ends in the same vertex, uses every vertex of G at least once, and uses every edge exactly edge exactly one.

10.2.4 Theorem 10.2.2

If a graph has an Euler circuit, then every vertex has even degree.

10.2.5 Theorem 10.2.3

if a graph G is connected and the degree of every vertex is even, then G has an Euler circuit.

10.2.6 Example 10.2.6

Insert graph here.